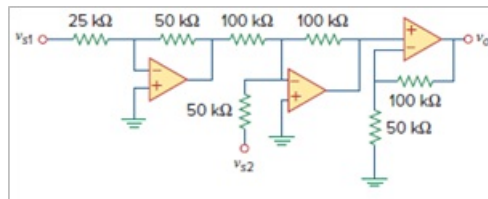


Problem

Find v_o in the op amp circuit of Fig. 5.84.

Figure. 5.84:



Step-by-step solution

Step 1 of 4

Refer to figure 5.84 given in the textbook.

Redraw the circuit with node voltages as shown in figure 1.

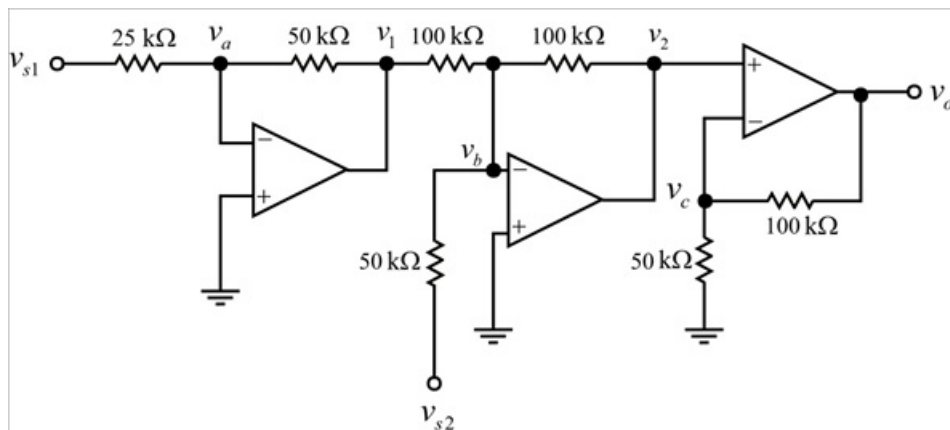


Figure 1

Step 2 of 4

Apply Nodal analysis at the inverting terminal of first op-amp with node voltage v_a .

$$\frac{v_a - v_{s1}}{25 \times 10^3} + \frac{v_a - v_1}{50 \times 10^3} = 0$$

Substitute 0 for v_a .

Since, both the inverting and non-inverting terminals of the op-amp are grounded virtually.

$$\frac{0 - v_{s1}}{25 \times 10^3} + \frac{0 - v_1}{50 \times 10^3} = 0$$

$$v_1 = (50 \times 10^3) \times \left(\frac{-v_{s1}}{25 \times 10^3} \right)$$

$$v_1 = -2v_{s1}$$

[Comments \(1\)](#)

☐

Anonymous

For the first part of the equation, shouldn't it be $v_{s1} - v_a$ in the numerator?

Step 3 of 4

Apply Nodal analysis at the inverting terminal of second op-amp with node voltage v_b .

$$\frac{v_b - v_{s2}}{50 \times 10^3} + \frac{v_b - v_1}{100 \times 10^3} + \frac{v_b - v_2}{100 \times 10^3} = 0$$

Substitute 0 for v_b .

Since, both the inverting and non-inverting terminals of the op-amp are grounded virtually.

$$\frac{0 - v_{s2}}{50 \times 10^3} + \frac{0 - v_1}{100 \times 10^3} + \frac{0 - v_2}{100 \times 10^3} = 0$$

Substitute $-2v_{s1}$ for v_1 .

$$\frac{0 - v_{s2}}{50 \times 10^3} + \frac{0 - (-2v_{s1})}{100 \times 10^3} + \frac{0 - v_2}{100 \times 10^3} = 0$$

$$\frac{0 - v_{s2}}{1} + \frac{0 - (-2v_{s1})}{2} + \frac{0 - v_2}{2} = 0$$

$$-v_{s2} + v_{s1} - \frac{v_2}{2} = 0$$

$$v_2 = 2(v_{s1} - v_{s2})$$

Step 4 of 4

Apply Nodal analysis at the inverting terminal of the third op-amp with node voltage v_c .

$$\frac{v_c}{50 \times 10^3} + \frac{v_c - v_o}{100 \times 10^3} = 0$$

Substitute v_2 for v_c .

Since, both the inverting and non-inverting terminals of the op-amp are grounded virtually.

$$\frac{v_2}{50 \times 10^3} + \frac{v_2 - v_o}{100 \times 10^3} = 0$$

$$\frac{v_o}{100 \times 10^3} = \frac{v_2}{50 \times 10^3} + \frac{v_2}{100 \times 10^3}$$

$$\frac{v_o}{2} = \frac{v_2}{1} + \frac{v_2}{2}$$

$$v_o = 3v_2$$

Substitute $2(v_{s1} - v_{s2})$ for v_2 .

$$v_o = 3 \times 2(v_{s1} - v_{s2})$$

$$= 6v_{s1} - 6v_{s2}$$

Therefore, the output voltage v_o is $\boxed{6v_{s1} - 6v_{s2}}$.